LAB 6

Introduction to **Object-Oriented Programming**

Overriding and Overloading

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Lab06 Exercises

❖ Lab06.1. Overriding methods of a Math class

Write a program to create a class “Math” and inherit from following “AbstractMath”
After all, override all functions and implement its own body statement to calculate accordingly.

*** You are not allowed to use existing libraries or classes in Java

```java
class AbstractMath {
    int reverseNumber(int num) {
        // override to return a reverse number
        // ex. 12345 to 54321
        return 0;
    }

    int roundedNumber(double num) {
        // override to return a arounded number
        // ex1. 3.14 to 3
        // ex2. 3.5 to 4
        // ex3. 3.7 to 4
        return 0;
    }

    int countDistinct(int[] values) {
        // override to return the count of unique value
        // ex1. { -1, -1, 1, 0, 1, 5, 9 } -> 3
        // ex2. { -5, 1, 0, 6, 5, 0 } -> 4
        return 0;
    }

    int decimalToBinary(int decimal) {
        // override to return the binary
        // coverted from decimal
        // ex1. 27 -> 11011
        // ex2. 121 -> 1111001
        return 0;
    }

}
```

Console:

```
==== Menu ===
1.  Reverse a number
2.  Rounded number
3.  Count distinct number
4.  Decimal to Binary
5.  Quit
Choose an opt: 1

::: Reverse a number :::
Input number: 4567
Output is: 7654
```

Lab06 Exercises

❖ Lab06.2. Overloading methods of a Math class

Write a program to create a math class with following return methods:

- sum (with 2 params)
- sum (with 3 params)
- sum (with array of value)
- multiply (with 2 params)
- multiply (with 3 params)
- multiply (with array of value)
- max (with 3 params)
- max (with 4 params)
- max (with array of value)
- min (with 3 params)
- min (with 4 params)
- min (with array of value)
- average (with 3 params)
- average (with 4 params)
- average (with array of value)

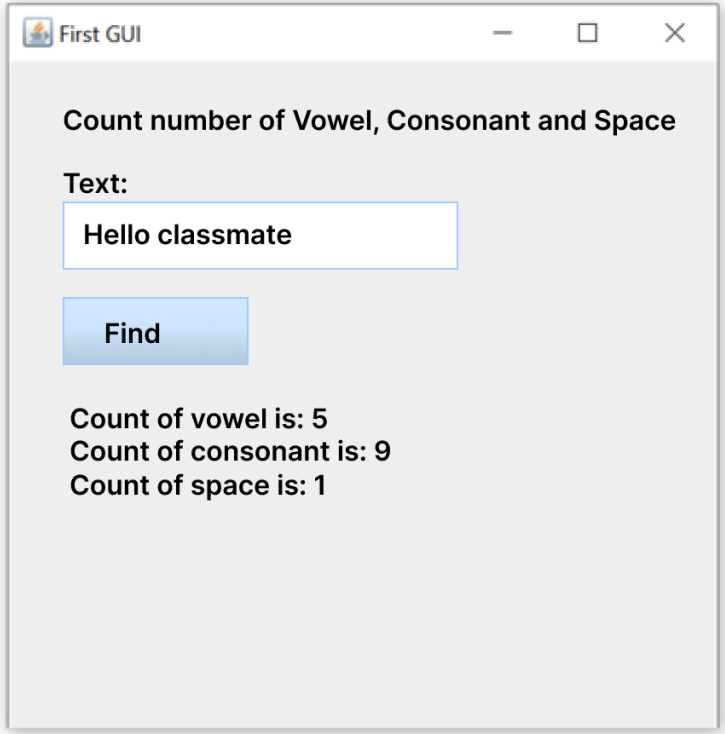
Console:

```
==== Menu ===
1.  Sum of two values
2.  Sum of three values
3.  Sum of many values
4.  Multiply of two values
5.  Multiply of three values
6.  Multiply of many values
7.  Max of two values
8.  Max of three values
9.  Max of many values
10. Min of two values
11. Min of three values
12. Min of many values
13. Average of two values
14. Average of three values
15. Average of many values
16. Quit
```

Lab06 Exercises

❖ Lab06.3. Character counter

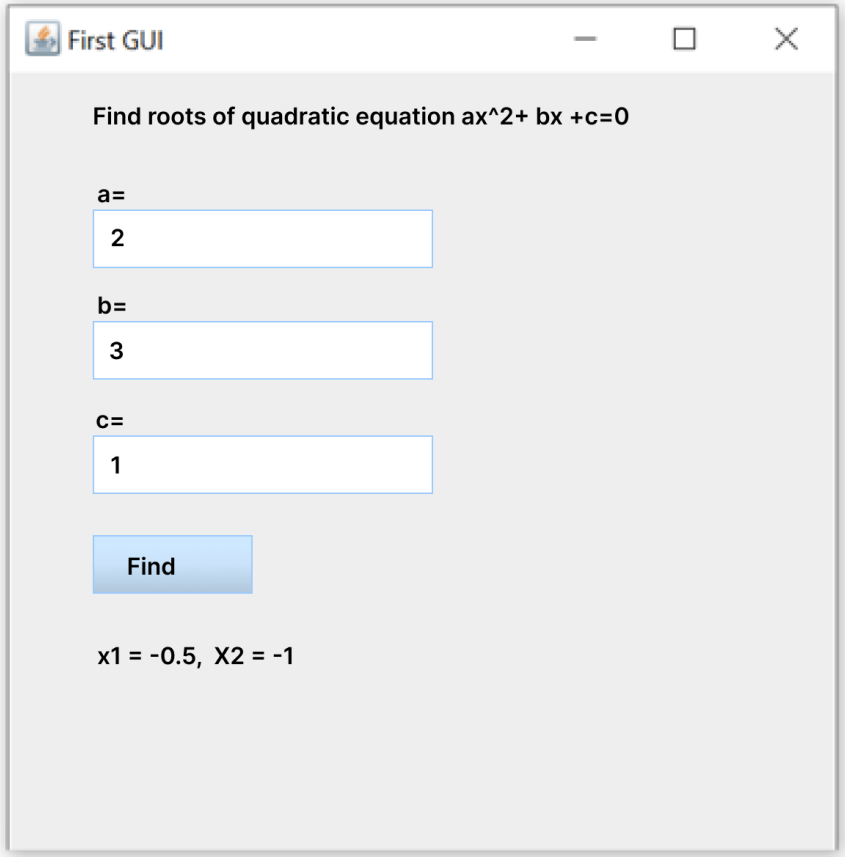
Write a program using to count number of Vowel, Consonant and Space:



Lab06 Exercises

❖ Lab06.4. Quadratic equation

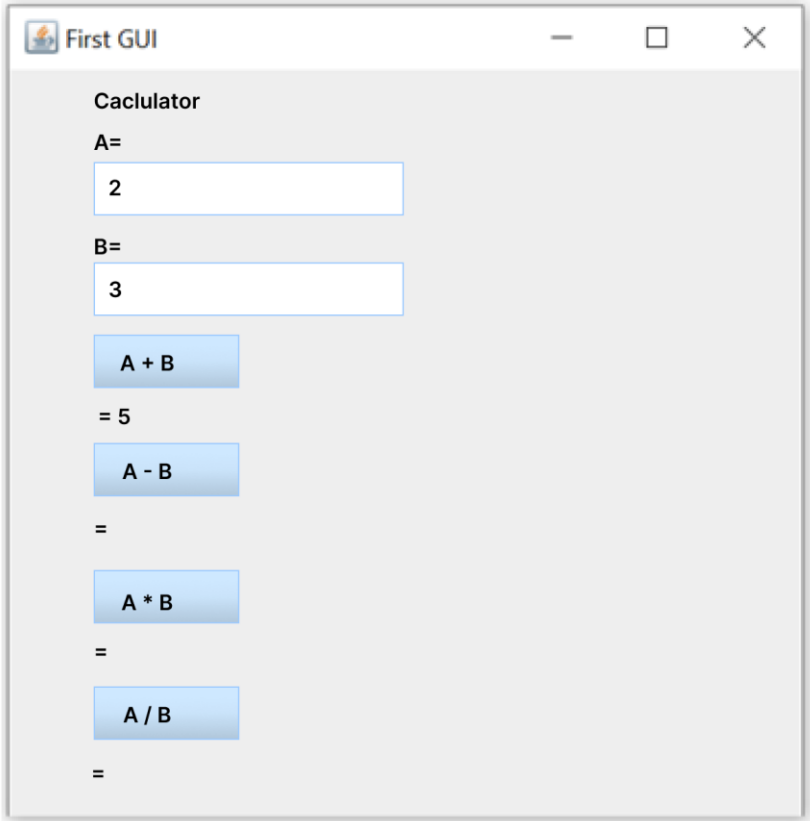
Write a program to find roots of quadratic equation ax^2+ bx +c=0 when “a”, “b” and “c” are given by user input



Lab06 Exercises

❖ Lab06.5. Simple Calculator

Write a program to perform calculation of two inputs with operators, addition, subtraction, multiplication and division



Brainstorming your final project

👬

A team with 3 two randomized students

Examples:

1. School Management System
2. Point of Sale (POS) System
3. Task management
4. Real Estate Sale System
5. Hotel Management System
6. Coffee Shop Management System
7. Human Resource System
8. Game
9. Etc.

4 weeks

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Good luck 👌

✋ How to submit my Lab works???

👉 Screenshot your source codes and answers as a single **pdf** file and submit to the **Moodle** with Java file attachment as well.

Note: Don’t zip the files for submission